LATEX

Date

Sets

collection of well defined and distinct objects.
سب کی same رائے ہو

Order does not matter in set.
can be unordered.

$a \in A \longrightarrow$ a is an element of set A.
↓ set name

Descriptive :

set of first five natural number.

Tabular :

$A = \{1, 2, 3, 4, 5\}$

Set Builder Notation :

$A = \{x \in N,\ 1 \le x \le 5\}$ / $\{x \in N,\ 1 \le x < 6\}$

$N = \{0, 1, 2 \ldots\ldots\}$ (Natural no).
$Z = \{\ldots\ldots -2, -1, 0, 1 \ldots\ldots\}$ (Integer)
$Z^+ = \{1, 2, 3 \ldots\ldots\}$ (+ve Integer).
$Q = \{p/q \mid p \in Z,\ q \in Z\ \&\ q \neq 0\}$ (rational no)
$R =$ (real no)
$R^+ =$ +ve real no.
$C =$ complex no (Real + imaginary)

نیچے 0 نہیں آنا چاہیے

Real numbers can not be written in Tabular form.

capable of becoming actual.
↓
Potential

In Q, we see potential position.
e.g $5 = \frac{5}{1}$] potential
actual

coll | a ~~person~~ number whose square is +ve is a real number. |

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- $\sqrt{2}$ cannot be written in $\frac{p}{q}$.

any square root which don't have complete square

(don't stop) · non terminating
(no pattern) · non recurring !] Irrational.

✓ pattern
✗ terminate

- Open & Close Intervals.

[2,5] closed interval, 2 and 5 are included
(2,5) open interval, 2 and 5 are not included

Equal Set:

$A = \{1, 2, 3\}$
$B = \{2, 3, 1\}$

$\{1, 3, 3, 3, 5, 5, 5\}$
$=$
$\{1, 3, 5\}$

Empty set: $\{\ \}$

Singleton set: $\{1\}$

Size of set: No. of elements in set

$|A| = 3$ for $A = \{1, 2, 3\}$

Subset:

set A is subset of B if every element of A is also an element of B.

$A \subseteq B$

$B = \{1, 2, 3\}$ $A \subseteq B$
$A = \{1, 2, 3\}$
$A \subseteq B$ $A \subset B$
$B \subseteq A$

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• ϕ is a subset of every set.

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~~Proper~~ Subset, Superset.

can or can not be equal.

$A = \{1, 2, 3, 4, 5\}$
$B = \{2, 3, 5\}$
$C = \{1, 2, 3, 4, 5\}$

$B \subseteq A$ ← B is a subset of A and A is superset of B.

$C \subseteq A$

$D = \{4, 9, 10\}$

$D \not\subseteq A$.

• $\subseteq$ can be proper & improper subset

• $\subset$ proper subset.

Proper Subset:

cannot be equal.

$A = \{1, 2, 3, 4, 5\}$
$B = \{3, 2, 1\}$

$B \subset A$

Improper Subset

$A = \{1, 2, 3, 4, 5\}$
$B = \{1, 2, 3, 4, 5\}$

$A = B$.

Cartesian Product

$A = \{1, 2, 3\}$
$B = \{a, b, c\}$

$A \times B = \{(1,a), (1,b), (1,c), (2,a), (2,b), (2,c), (3,a), (3,b), (3,c)\}$.

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$2 \times 2 \times 2 \times 2 = 10$

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$2^3 = 8$

$2^3 = 8$

(1,2) (2,1)

Power Set:

Subsets of given set in a single set.

e.g:

Q1 $A = \{1, 2, 3\}$

$P(A) = \{ \phi, \{1\}, \{2\}, \{3\}, \{1,2\}, \{1,3\}, \{2,3\}, \{1,2,3\} / A \}$.

$\rightarrow$ Number of elements : $2^n = 2^3 = 8$

Q2 $A = \{1, \{2,3\}\}$

$2^2 = 4$

$P(A) = \{ \phi, \{1\}, \{\{2,3\}\}, \{1, \{2,3\}\} \}$

Q3 $A = \{1, 2, \phi, \{\phi\}\}$

$2^4 = 16$.

$P(A) = \{ \phi, \{1\}, \{2\}, \{\phi\}, \{\{\phi\}\},$

$= \{\phi, 1\}, \{\phi, 2\}, \{\phi, \phi\}, \{\phi, \{\phi\}\},$

$\{1, \phi\}, \{2, \phi\} \{1, 2\}, \{1, \phi\}, \{1, \{\phi\}\}, \{2, \phi\},$

$\{2, 1\}, \{2, \phi\}, \{2, \{\phi\}\}, \{\phi, \phi\}, \{\phi, 1\}, \{\phi, 2\}, \{\phi, \{\phi\}\},$

$\{\{\phi\}, \phi\}, \{\{\phi\}, 1\}, \{\{\phi\}, 2\}, \{\{\phi\}, \phi\}$

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Exercises: Q21, 23, 24, 28, 7, 8, 9
(sets).

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$P(A) = \{\phi, \{1\}, \{2\}, \{\phi\}, \{\{\phi\}\}, \{1,2\}, \{1,\phi\},$
$\{1,\{\phi\}\}, \{2,\phi\}, \{2,\{\phi\}\}, \{\phi,\{\phi\}\},$
$\{1,2,\phi\}, \{1,2,\{\phi\}\}, \{2,\phi,\{\phi\}\},$
$\{1,\phi,\{\phi\}\}, \{1,2,\phi,\{\phi\}\}$

Q4 $A = \{\{1,2,3,4\}\}$

$2^1 = 2$

$P(A) = \{\phi, \{\{1,2,3,4\}\}\}$

Q9
a) No
b) No

- $\in$ member
- $\subset$ proper subset
- $\subseteq$ subset.

Exersice:

Q7
a) Yes
b) No
c) Yes $\{2, \{2\}\}$
d) No $\{\{2\}, \{\{2\}\}\}$
e) No
f) No.

$\{0\} \subset \{5, \{0\}\}$ Yes

Q8 by your own

f) $\{0\} \subset \{0\}$

No.

g) $\{0\} \subseteq \{0\}$

Yes

A set is always a subset of itself

a) $0 \in \phi$
$0 \in \{\}$ (No)

b) $\phi \in \{0\}$
No, if subset $\subseteq$ then Yes.

c) $\{0\} \subset \phi$
No

d) $\phi \subset \{0\}$

Yes

e) $\{0\} \in \{0\}$.

~~Yes~~ No. , $0 \in \{0\}$ Yes

$\{0\} \in \{\{0\}\}$ Yes

Union $A \vee B$ $A \cup B$ $\{x \mid x \in A \vee x \in B\}$

$\cup$ $\vee$

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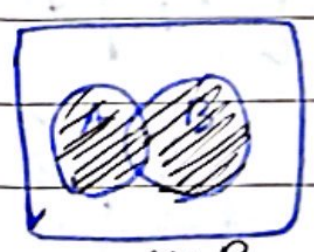
Q8 a) No
b) No
c) Yes
d) Yes
e) Yes
f) No

Union:



$A \cup B$ AUB

$A \cup B$, $A = \{1, 2, 3, 4\}$, $B = \{3, 4, 6, 7\}$

$A \cup B = \{1, 2, 3, 4, 6, 7\}$

OR

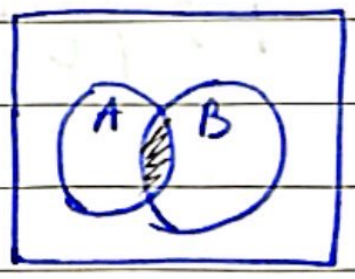
$A \cup B = \{x \mid x \in A \vee x \in B\}$

Intersection / Overlapping: (common)

$A \cap B = \{x \mid x \in A \wedge x \in B\}$

AND

$A \cap B = \{3, 4\}$



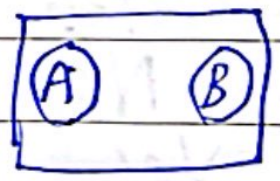
$A \cap B$. $\wedge$

Disjoint Sets: (No common)

(intersection empty set)

$A \cap B = \phi$

$A = \{1, 2, 3\}$, $B = \{4, 5, 6\}$



$A = \{1, 2, 3\}$
$B = \{4, 5, 6\}$

$$\boxed{|A \cup B| = |A| + |B| - |A \cap B|}$$

cardinality

size / no. of elements.

$A = \{1, 2, 3, 4\}$, $B = \{3, 4, 6, 7\}$

$A \cup B = \{1, 2, 3, 4, 6, 7\}$

$A \cup B = |A| + |B| - |A \cap B|$

$6 = 4 + 4 - 2$

$6 = 6$

complete / دو سیٹس کو مکمل کرے

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Difference of Sets (A − B) / (Complement of B with respect to A)

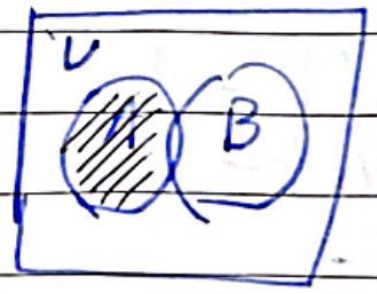
→ elements of A write

↳ B elements not written which are in A.

$$A - B = \{x \mid x \in A \wedge x \notin B\}$$

$A = \{1, 2, 3, 4\}$, $B = \{3, 4, 6, 7\}$

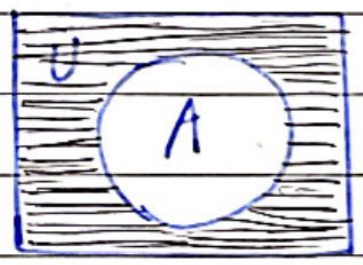
$A - B = \{1, 2\}$



A − B

$A = \{2, 4, 6\}$

$A' = U - A = \{1, 3, 5, 7, 8, 9, 10\}$



$$\bar{A} = \{x \in U \mid x \in A\}$$

↓

A complement

Set Identities :

a ya b ki koi bhi value put

ku dain to L.H.S = R.H.S

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$A \cap U = A$
$A \cup \phi = A$ — Identity law

$A \cup U = U$
$A \cap \phi = \phi$ — Domination law

$A \cup A = A$
$A \cap A = A$ — Idempotent law

$\overline{(\bar{A})} = A$ — Complementation law

$A \cup B = B \cup A$
$A \cap B = B \cap A$ — Commutative law

$A \cup (B \cup C) = (A \cup B) \cup C$
$A \cap (B \cap C) = (A \cap B) \cap C$ — Assosiative law

$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ — Distributive law

$\overline{A \cap B} = \bar{A} \cup \bar{B}$
$\overline{A \cup B} = \bar{A} \cap \bar{B}$ — Demorgan's law

$A \cup (A \cap B) = A$
$A \cap (A \cup B) = A$ — Absorption law

$A \cup \bar{A} = U$
$A \cap \bar{A} = \phi$ — Complement law

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$\sim A \cap B$

Example 11:

$\overline{A \cap B} = \bar{A} \cup \bar{B}$

$\overline{A \cap B} = \{x \mid x \notin A \cap B\}$

$= \{x \mid \sim(x \in (A \cap B))\}$

$= \{x \mid \sim(x \in A \wedge x \in B)\}$

$= \{x \mid \sim(x \in A) \vee \sim(x \in B)\}$

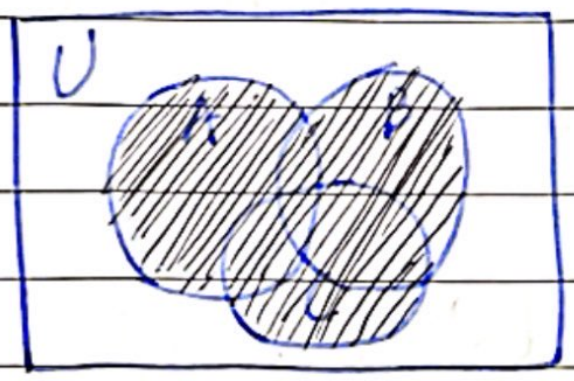
$= \{x \mid x \notin A \vee x \notin B\}$

$= \{x \mid x \in \bar{A} \vee x \in \bar{B}\}$

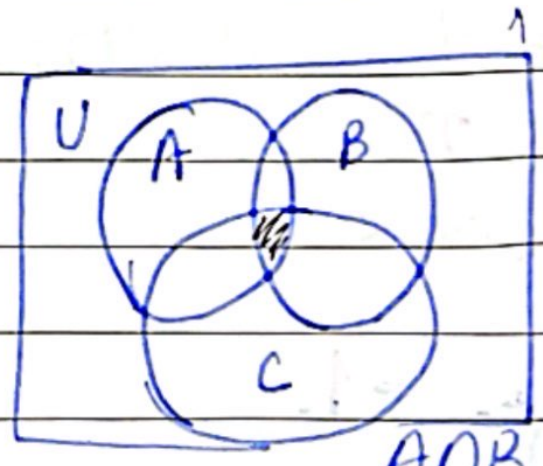
$= \{x \mid x \in \bar{A} \cup \bar{B}\} = \bar{A} \cup \bar{B}$

Generalized Unions & Intersections

more than 2



$A \cup B$



$A \cap B$

Union of Collection of Sets:

$A_1 \cup A_2 \cup \ldots A_n = \bigcup_{i=1}^{n} A_i \rightarrow$ will change

ending index

starting index

Intersection of Collection of sets

$A_1 \cap A_2 \cap \ldots A_n = \bigcap_{i=1}^{n} A_i$

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Golden ratio.

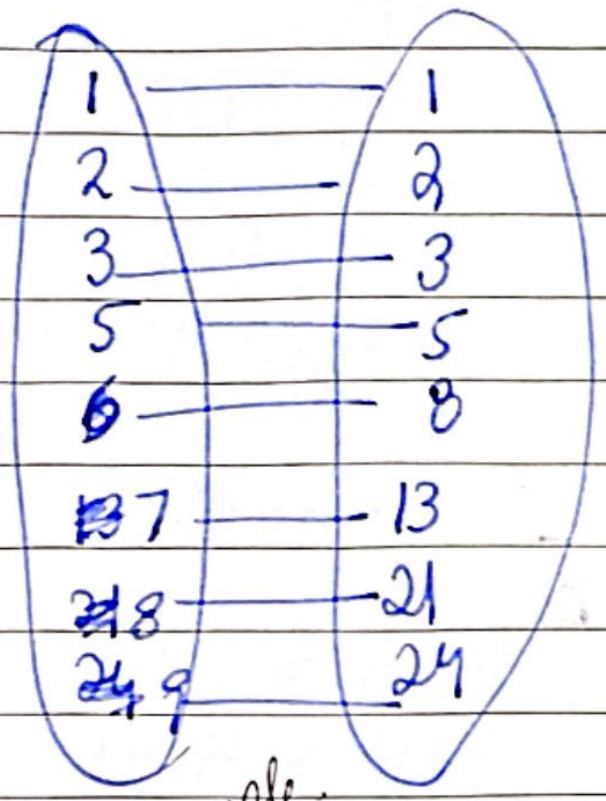
Date 2.4 Sequences & Summations.

Sequence:
→ a function.

1, 2, 3, 5, 8, 13, 21, 34 ....

$a_n \longrightarrow$ one element

$[a_n] \rightarrow$ all generated elements



Example 1.

$[a_n]$, $a_n = \frac{1}{n}$, $a_1, a_2, a_3, a_4, \ldots$ starts with

where.

$1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \ldots$

$a_n = \frac{1}{1} = 1$

$a_n = \frac{1}{2} = \frac{1}{2}$

$a_n = \frac{1}{3}$

$a_n = \frac{1}{4}$

$\vdots$

2.4 Exercise.

Q1. $[a_n]$ where $a_n = 2 \cdot (-3)^n + 5^n$

a) a_0

$= 2 \cdot (-3)^0 + 5^0$

$= 2 + 1 = 3$.

Q2, Q3, Q4

1.4

$2\sqrt{3}$

2/10

Q1-8, 9, 10, 11

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Q5 a) 2, 5, 8, 11, 14, 17, 20, 23, 26, 29

b) 1, 1, 1, 2, 2, 2, 3, 3, 3, 4, 4, 4, 5, 5, 5, 6, 6, 6, ...

c) 1, 1, 3, 3, 5, 5, 7, 7, 9, 9 . . . .

recursive function

jo function khud ko call krai.

Q9 a) $a_n = 6a_{n-1}$, $a_0 = 2$. (starting point).

Yahan n ki value aisai put krni hai k ao ~~ao~~ aa jaye.

$a_1 = 6a_{1-1}$

$a_1 = 6a_0$

$a_1 = 6 \cdot 2$

$a_1 = 12.$

$a_2 = 6_{a_1}$

$a_2 = 6(12)$

$= 72$

$12 \times 6 = 72$

$72 \times 6 = 432$

$432 \times 6 = 2592$

$a_3 = 6a_2$
$= 6(72)$
$= 432$

$a_4 = 6a_3$
$= 6(432)$
$= 2592$

$a_5 = 6a_4$
$= 6(2592)$

c) $a_n = a_{n-1} + 3a_{n-2}$ $a_0 = 1$, $a_1 = 2$

$a_2 = a_1 + 3a_0$
$= 2 + 3(1)$
$= 5$

Sequence → Arithematic sequence
→ Geometric sequence.

Arithematic Sequence:

d = common difference.

$$\boxed{a_n = a_1 + (n-1)d}$$

Total no. of terms

common difference

formula

e.g: 5, 9, 13, 17 . . . . . .

Find 100th term.

Find any value of arithematic progression

$= 5 + (100-1) \times 4$

$= 5 + 99 \times 4 = 401$

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Sum of first n terms of sequence -

$$\boxed{S_n = \frac{n}{2}\{2a_1 + (n-1)d\}}$$

$$= \frac{100}{2}\{2.5 + (100-1)4\}$$

$$= 50(10+396) = 50(406) = 20300$$

Geometric Sequence.

difference matters. → have common ratio b/w two consecutive terms.

2, 6, 18, 54, ...

$$= 3 = \frac{6}{2} = \frac{18}{6} = \frac{54}{18}$$

$$\boxed{a_n = a\,r^{n-1}}$$ (formula to find n^{th} term)

To find 20th value :

$$a_{20} = 2.3^{20-1}$$

$$a_{20} = 2.3^{19}$$

$$\boxed{S_n = \frac{a(r^n - 1)}{r-1}}$$ (to find sum)

$$= \frac{2(3^{20}-1)}{3-1} = \frac{2}{2} \times (3^{20}-1)$$

$$= 3^{20} - 1$$

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CHESS, PERSON, KING

5, 5, 5, 5 ....
this is both A.P & G.P.

8 چاول = ایک
96 رتی = 1 تولہ
85 تولہ = 1 کلوگرام

8 x 96 x 85 x
40 x 25

1 + 2 + 4 + 8 + ... 64 term

$$S_{64} = \frac{1(2^{64}-1)}{2-1} = 2^{64}-1$$

$$S = \frac{a(r^n-1)}{r-1}$$

Summations :

sigma (sum) $\sum_{i=1}^{5}$ i — expression

5 — ending value

Starting value

$$\sum_{i=1}^{5} i = 1+2+3+4+5 = 15$$

$$\sum_{i=1}^{3} 2i = 2+4+6 = 12.$$

$$\sum_{i=1}^{x} i = 1+2+3+ \ldots x.$$ (This is A.P) so,

$$S_n = \left[\frac{n}{2} \quad 2a_1 + (n-1)d\right]$$

$$\sum_{i=1}^{50} 5 = 5+5+5 \ldots 5$$
$$= 5 \times 50 = 250$$

$$\sum_{i=5}^{15} 1 = \overset{5}{1}+\overset{6}{1}+\overset{7}{1}+\overset{8}{1}+\overset{9}{1}+\overset{10}{1}+\overset{11}{1}+\overset{12}{1}+\overset{13}{1}+\overset{14}{1}+\overset{15}{1}$$
$$= 11.$$

5 sai 15 tk
1 add ho
rha.

$15 - 5 + 1 = 11$

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$$\sum_{i=a}^{b} 2 = 2(b-a+1)$$

Double Summation

Example : 21.

$$\sum_{i=1}^{4} \sum_{j=1}^{3} ij$$

$$\sum_{i=1}^{4} (i+2i+3i)$$

$$\sum_{i=1}^{4} (6i) = 6(1+2+3+4)$$
$$= 60$$

$$\sum_{i=0}^{x} \sum_{j=0}^{y} \sum_{k=0}^{z} 1$$

$$= \sum_{i=0}^{x} \sum_{j=0}^{y} z+1$$

z is constant w.r.t j.

$$= (z+1) + (z+1) + (z+1) \ldots$$

$$= \sum_{i=0}^{x} (y+1)(z+1)$$

$$= (x+1)(y+1)(z+1)$$

Formula for sum of squares

$$n^2 = \frac{n(n+1)(2n+1)}{6}$$

Sum of cubes.

$$n^3 = \frac{n^2(n+1)^2}{4}$$

value of k

$$\frac{n(n+1)}{2}$$

$$\sum_{i \in (3,7,11)} i$$

$$3+7+11 = 21.$$

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Find $\sum_{K=50}^{100} K^2$ if $K=1$ we use $\frac{n(n+1)(2n+1)}{6}$ so,

$$\sum_{K=1}^{100} K^2 = \sum_{K=1}^{49} K^2 + \sum_{K=50}^{100} K^2$$

$$\sum_{K=1}^{100} K^2 - \sum_{K=1}^{49} K^2 = \sum_{K=50}^{100} K^2$$

$$\frac{100(101)(201)}{6} - \frac{49(50)(99)}{6} = \sum_{K=50}^{100} K^2$$

Exercise:
Q25

EMP

Mathematical Induction (Chp # 5)

proof technique.

e.g:

Sum of n natural no. : $\frac{n(n+1)}{2}$

Proof that wheather it this formula is correct or not.

Mathematical Induction,

To 1) check $n=1$

base case.

2) ~~Ch~~ Put K

3) Put K+1.

Formula

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$\therefore$ D is not a +ve integer.

$\star \dfrac{n(n+1)}{2}$

- Put $n = 1$

$1 = \dfrac{1(1+1)}{2} = \dfrac{2}{2}$ #

$1 = 1$ (True)

- Put K

$\dfrac{K(K+1)}{2} + K + 1 = \dfrac{K^2+K+2K+2}{2} = \dfrac{K^2+3K+2}{2}$

Put next term $= \dfrac{(K+1)(K+2)}{2}$

- Put K+1 $= \dfrac{(K+1)^2(K+1+1)}{2}$.

$\dfrac{(K+1)(K+1+1)}{2}$.

$\star$ $n^3 - n$ (divisible by 3 formula). (Without Mathematical Induction)

$n(n^2-1)$

$n(n+1)(n-1)$

smallest $(n-1)(n)(n+1)$

here any one should be ÷ble by 3.

456

678

8910

Use mathematical induction to prove that

$1 + 2 + 2^2 + \ldots\ldots 2^n = 2^{n+1} - 1$ for all $\underbrace{\text{non-ve integers}}_{n}$.

$2^0 + 2^1 + 2^2 + 2^3 \ldots\ldots 2^n = 2^{n+1} - 1$, (0 is included)

- $2^0 = 2^{0+1} + 1$

$1 = 2 - 1$

$1 = 1$

$2^0 + 2^1 = 2^{1+1} - 1$

$1 + 2 = 2^2 - 1$

$3 = 3$

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Put $n = k$.

$1 + 2 + 2^2 + 2^3 + \ldots\ldots 2^n = 2^{n+1} - 1$

$\cdot\ 1 + 2 + 2^2 + 2^3 \ldots\ldots 2^k + 2^{k+1} = 2^{k+1} - 1 + 2^{k+1}$ — next term

$= 2^{k+2} - 1$

$= 2^{k+1+1} - 1$ — (i)

$2^{n+1} - 1$

put $k+1$ in n

$2^{k+1+1} - 1$ — (ii)

(i) and (ii) proved.

Ex: 02:

Conjecture a formula for the sum of the first n +ve odd integer. Then prove your conjecture using mathematical induction.

$1 = 1$

$1 + 3 = 4$

$1 + 3 + 5 = 9$

$1 + 3 + 5 + 7 = 16$

$1 + 3 + 5 + 7 + 9 = 25$

$a_n = a + (n-1)d$

$2n - 1 = 1 + (n-1)2$

$1, 3, 5, 9, 16, 25 \ldots\ldots 2n - 1 = n^2$

Put $n = 1$

$1 = 1^2$

$1 = 1$

Put $n = k$

$1 + 3 + 5 + \ldots\ldots 2k - 1 = k^2$

$1 + 3 + 5 + \ldots\ldots 2k - 1 + 2 = k^2 + 2k + 1$

$= (k+1)^2$

$1 + 3 + 5 + \ldots\ldots 2k + 1$

$n^2 = (k+1)^2$

because all numbers are increasing by 2.

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Show that if $2^k < k!$ for an arbitary +ve integer k when $k \geq 4$, then $2^{k+1} < (k+1)!$

$2^k < k!$ let,

let $2^4 < 4! = 16 < 24$

4
3
12
8
24

• Put $n = 4$

$2^4 < 4!$

$16 < 24$ (True)

• Put $n = k$

$2^k < k!$

$2^{k+1} = 2 \cdot 2^k$

$2^{k+1} < 2 \cdot k!$

$2^{k+1} < (k+1) \cdot k!$ because we want $k+1$ in answer, so we write this.

$2^{k+1} < (k+1)!$

$\boxed{2^{k+1} < (k+1)!}$ we have to prove this.

$6 \cdot 5! = 6!$

$7 \cdot 6! = 7!$

Example: 09

Prove that $7^{n+2} + 8^{2n+1}$ is divisible by 57 for every non-ve integer n.

$= 7^{n+2} + 8^{2n+1}$

Put $n = 0$

$= 7^{0+2} + 8^{0+1} = 49 + 8 = 57$

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~~Put any +ve integer.~~

~~let $n = 3$~~

~~$7^{3+2} + 8^{2(3)+1}$~~

~~$7^5 + 8^7 = 211397$~~

~~Put $n = 0$~~ Put $n = K$

• Put $n = K$

$7^{K+2} + 8^{2K+1}$

• Put $K+1$

$7^{K+1+2} + 8^{2(K+1)+1}$

$7^{K+3} + 8^{2K+3}$

$7 \cdot 7^{K+2} + 8^2 \cdot 8^{2K+1}$

$7 \cdot 7^{K+2} + 64 \cdot 8^{2K+1}$

$7 \cdot 7^{K+2} + (7+57) \cdot 8^{2K+1}$

$7 \cdot 7^{K+2} + 7 \cdot 8^{2K+1} + 57 \cdot 8^{2K+1}$

$7(7^{K+2} + 8^{2K+1}) + 57 \cdot 8^{2K+1}$

$\underbrace{7^{n+1+2} + 8^{2(K+1)+1}}$ } we have to Proof this

we have to prove 57 can divide it.

As this side and this side is ÷ble by 57 so prove.

Exercise:

Q7 Prove that $3 + 3 \cdot 5^1 + 3 \cdot 5^2 + \dots + 3 \cdot 5^n = 3(5^{n+1} - 1)/4$

where n is a non-ve integer.

Base case: Put $n = 0$;

$3 \cdot 5^0 + 3 \cdot 5^1 + 3 \cdot 5^2 + \dots 3 \cdot 5^n = 3(5^{n+1} - 1)/4$

$3 \cdot 5^0 = 3(5^1 - 1)/4$

$3 = 3(4)/4$

$3 = 3$

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Decimal.

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$3\cdot5^0 + 3\cdot5^1 = 3(5^{n+1}-1)/4$

$3 + 15 = 3(5^2-1)/4$

$18 = \frac{3(24)}{4}\ \frac{126}{21}$

$18 = 18$

$n = k.$

$3\cdot5^0 + 3\cdot5^1 + 3\cdot5^2 + \dots 3\cdot5^n = 3(5^{n+1}-1)/4$

$3\cdot5^0 + 3\cdot5^1 + 3\cdot5^2 + \dots 3\cdot5^k = 3(5^{k+1}-1)/4$

$n = k+1$

$3\cdot5^0 + 3\cdot5^1 + 3\cdot5^2 + \dots 3\cdot5^k + 3\cdot5^{k+1} = \left[3(5^{k+1}-1)/4\right] + 3\cdot5^{k+1}$

$3(5^{k+1}-1)/4 + 3\cdot5^{k+1} = 3(5^{k+1}-1)/4 + 3\cdot5^{k+1}$

$\cancel{3(5^{k+1+1}-1)/4 = 3(5^{k+1+1}-1)/4}.$

$= 3(5^{k+1}-1)/4 + 3\cdot5^{k+1}$

$= \frac{3\cdot5^{k+1} - 3 + 12\cdot5^{k+1}}{4}$

$= \frac{5^{k+1}(3+12) - 3}{4}$

$= \frac{5^{k+1}(15) - 3}{4}$

$\frac{3(5^{n+1}-1)}{4} = \frac{3(5^{k+1+1}-1)}{4}$

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$$\frac{15 \cdot 5^{k+1} - 3}{4}$$

$$\frac{3(5^1 \cdot 5^{k+1} - 1)}{}$$

$$\frac{3(5^{4k+2} - 1)}{4} = \frac{3(5^{k+1+1} - 1)}{4}$$

Q6 Prove that

$1 \cdot 1! + 2 \cdot 2! + \ldots\ldots n \cdot n! = (n+1)! - 1$ whenever n is a +ve integer: $n \geq 1$.

$n = 1$

$1 \cdot 1! = (1+1)! - 1$

$1 = 2! - 1$

$1 = 1$

$n = 2$

$2 \cdot 2! =$

4

$(2+1)!$

$6 - 1 = 5$

$3 \cdot 3! = 18$

$4! - 1$

$4 \times 3 \times 2$

$24 - 1$

Put $n = 1$

$1 \cdot 1! = (1+1)! - 1$

$1! = 2! - 1$

$1 = 1$

Put $n = k$:

$1 \cdot 1! + 2 \cdot 2! + 3 \cdot 3! + \ldots\ldots k \cdot k! = (k+1)! - 1$

$1 \cdot 1! + 2 \cdot 2! + \ldots\ldots\ldots\ldots k \cdot k! + (k+1) \cdot (k+1)! = (k+1)! - 1 + (k+1) \cdot (k+$

$= (k+1)! [1 + k + 1] - 1$

$= (k+1)! \cdot (k+2) - 1$

$(k+2)! - 1$

$= (k+1+1)! - 1$

$2 \times 2 \times 2 \times 2 \times 2 = 16$
$\frac{16}{2}$
32

Pg : 330.

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Q 21 Prove that $2^n > n^2$ if n is an integer greater than 4.

$n=1$

$2^5 > 5^2$

$32 > 25$ (True)

$n=K$

$2^K > K^2$

$n = K+1$

$\cancel{2^K + 2^{K+1}} > \cancel{(2K+1)^2}$

$2^K > K^2$

$\times$ by 2

$2^K \cdot 2 > K^2 \cdot 2$

$2^{K+1} > 2K^2$

$2^{K+1} > K^2 + K^2$

$2^{K+1} > K^2 + 2K + 1$

$\boxed{2^{K+1} > (K+1)^2}$

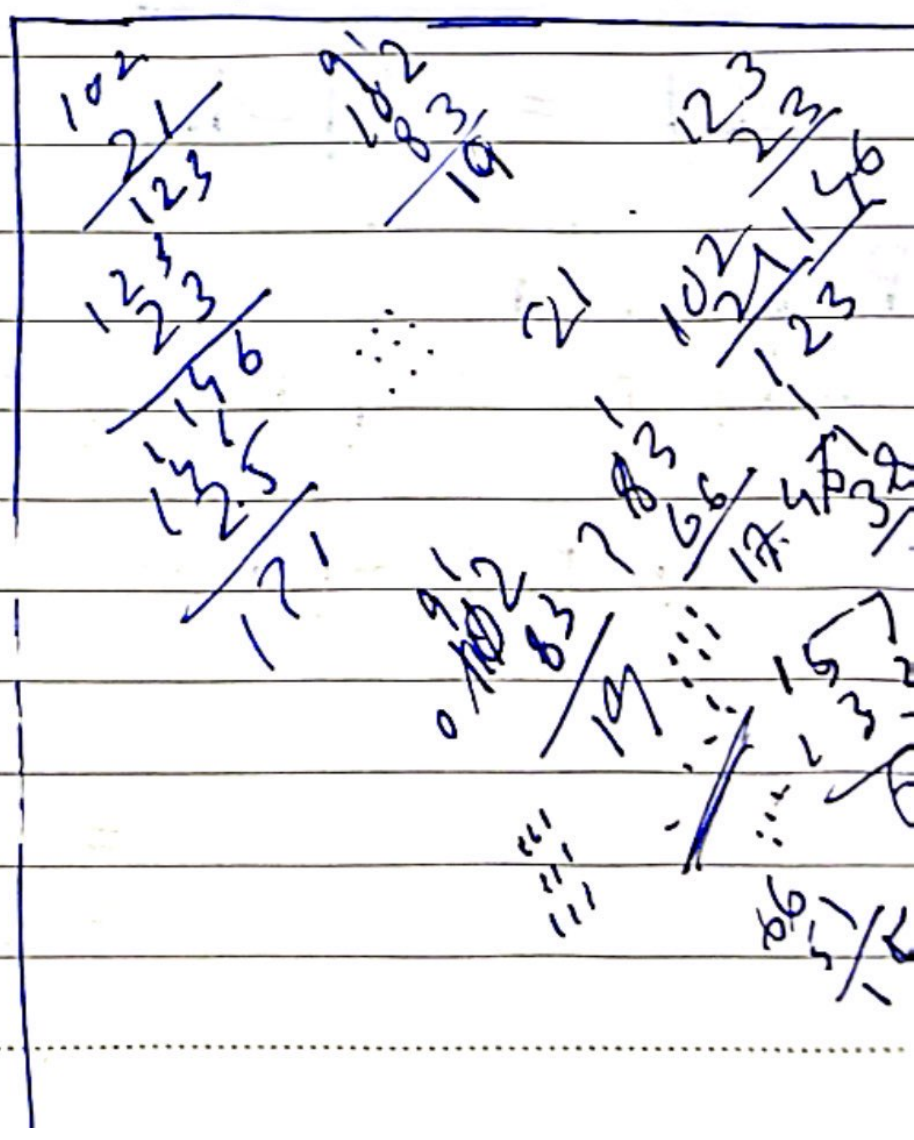
$\therefore 2^{K+1} > (K+1)^2$

we have to make this

$2K+1 < K^2$

for

$K < 4$



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Counting

Combinatorics : science of making combinations.
Enumeration : counting of objects with certain properties

Rules Of ~~Com~~ Counting :

→ Product Rule . $(n_1 \cdot n_2)$
→ Addition (Sum) Rule . $(n_1 + n_3)$

Product Rule:

Example 01 : (Basic Counting Principle)

1 2 3 4 5 6 7 8 9 10 11 12

$12 \times 11 = 132$.

for first person

after 1 person 11 are left.

$$\begin{array}{r} 12 \\ +12 \\ \hline 24 \\ 12\times \\ \hline 144 \\ -12 \\ \hline 132 \end{array}$$

Example 02.

A1, A2, ... A100
B1, B2, ... B100
⋮
Z1, Z2, ... Z100

2600

Sum Rule:

Example 01.

$37 + 83 = 120$

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Exercise :

Q7 How many different 3 letter initial can people have?

26^3

Q8 How many different three letter initials with non of letters repeated can people have?

$26 \cdot 25 \cdot 24$

Q9 How many different 3 letter initials are there that begin with an A?

26^2 $\underline{A}$ $\underline{26}$ $\underline{26}$.

fix

when anything is fix that power finishes.

Q10 How many bit strings are there of length 8?

2^8 $\underline{2}$ $\underline{2}$ $\underline{2}$ $\underline{2}$ $\underline{2}$ $\underline{2}$ $\underline{2}$ $\underline{2}$ choices

(0,1) (0,1) ...

Q11 How many bit strings of length 10 both begin & end with 1?

2^8

Date

Q13 How many bit strings with length not exceeding n, where n is a +ve integer, consist entirely of 1's not counting empty strings?

let, $n = 4$

Space.

Ans $= n$.

Q16. How many strings are there of form lowercase letters that have letter x in them?

$\underbrace{26}^{4} - \underbrace{25}^{4}$

سارے alphabets

x کے بغیر

جس میں x نہیں ہیں

Q47 In how many ways can a photographer at a wedding arrange six people in a row including bride & groom if

a) the bride must be next to groom. factorial (!)

B G _ _ _ _

~~6!~~ $10 \cdot 4!$

B G _ _ _ _

_ B G _ _ _

_ _ B G _ _

_ _ _ B G _

_ _ _ _ B G

5

when G is at first

5

$5+5 = 10$

b) bride must not be next to groom.

$6! - 10 \cdot 4!$

Ex: 6.1
Q1-40

Date. ~~The Pigeonhole Principle~~

Inclusion | Exclusion (Subtraction Rule)

Example 18.

How many bit strings of length 8 either start with 1 bit or end with two bits 00?

e.g. 11111100 ✓
11111011 ✓
10011100 ✗

fix 1 so ignore _ _ _ _ _ _ _ $= 2^7$

_ _ _ _ _ _ fix 00 $= 2^6$

1 _ _ _ _ _ _ 00 $= 2^5$

$2^7 + 2^6 - 2^5$

$128 + 64 - 32 = 160$

$|A_1 \cup A_2| = |A_1| + |A_2| - |A_1 \cap A_2|$

Permutation &	Combinations
order matters	order not matters.

If n is a +ve integer with $1 \le r \le n$,

$P(n, r) = n(n-1)(n-2)\ldots\underbrace{(n-r+1)}$

Total, chosen

if $n = 10$
$r = 7$
$10 \cdot 9 \cdot 8 \cdot 7 \cdot 6 \cdot 5 \cdot 4$
$10 - 7 + 1 = 4$

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$^nP_r = \frac{n!}{(n-r)!}$

Combinatorial explosion.

Combination.

order not matters.

$^nC_r = \frac{n!}{(n-r)!\,r!}$

Exercise : Q1 - 38

Q2 7!

Q1 $\{a, b, c\} = \{a, b, c\}, \{a, c, b\}, \{b, a, c\},$
$\{b, c, a\}, \{c, a, b\}, \{c, b, a\}$

Q10 6!

Q11 a) $^nC_r = {}^{10}C_4 = 210$ (exactly 4 1's)

global values

local values

b) at most 4 1's
c) ~~at least 4 1's~~ } range. $\frac{10!}{4!\cdot 6!} + \frac{10!}{3!\cdot 7!} + \frac{10!}{2!\cdot 8!} + \frac{10!}{1!\cdot 9!} + \frac{10!}{10!}$

$\frac{10!}{4!\,6!}$

$\frac{10\cdot 9\cdot 8\cdot 7\cdot}{4\cdot 3\cdot 2}$

$\frac{10!}{4!\,6!}$

$\frac{10\cdot 9\cdot 8\cdot 7\cdot 6!}{4!\,6!}$

$\frac{10\cdot 9\cdot 8\cdot 7}{4\cdot 3\cdot 2}$

$\cancel{x \geq 5}$ $(x \leq 5)$ زیادہ سے زیادہ : atmost

$(x \geq 5)$ کم سے کم : at least

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c) at least 4 1s (range)

$= \cancel{\frac{10!}{4!\,6!}}$

Pigeon Hole Principle

Decimal : 0 – 9 (10 digits)

Example : 03

0 – 100

101 scores

102 students.

Generalized Pigeon Hole Principle (Multiplicity version)

53 ⟶ 3

A ⟶ 1 time

B

C

⋮

Z

A ⟶ 2nd time

B

C

⋮

Z

A/B/C/D... ⟶ 3rd time.

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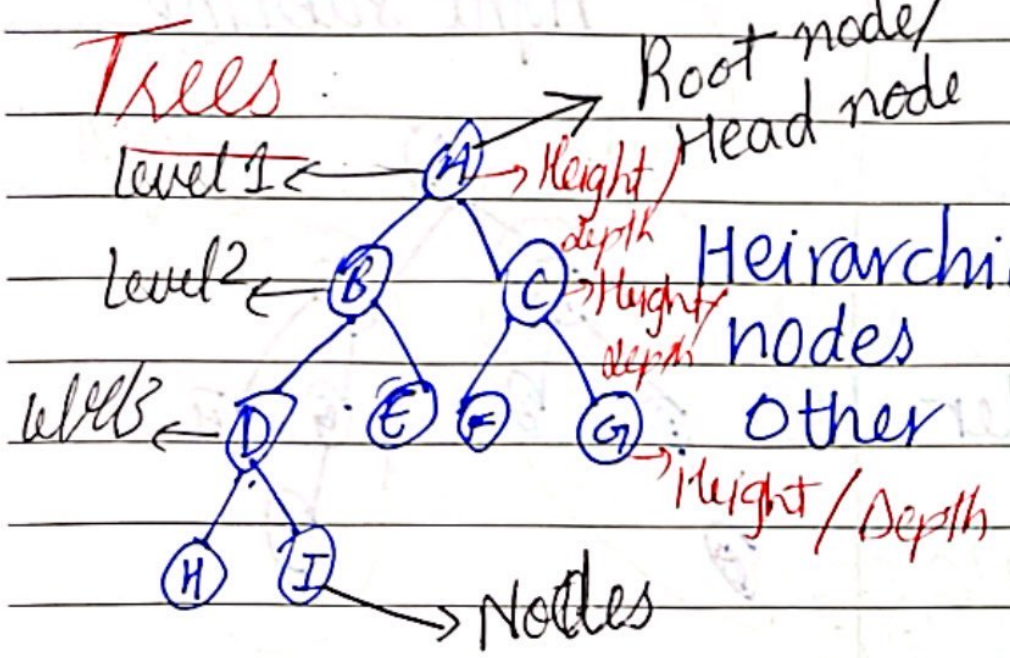
boxes	letters → 26
objects	words → 53

take cieling: $\left\lceil \frac{\text{Object}}{\text{boxes}} \right\rceil = \left\lceil \frac{N}{K} \right\rceil$ (Generalized Pigeon Hole principle)

Trees:



Heirarchial data structure having nodes consisting of relation with other nodes.

leaf node : has no child
Siblings : B, C are siblings
Cousins : F, E are cousins.

Binary tree : Tree whose node has maximum two children.

~~to wi~~

Traversing Binary Trees:

→ Pre order traversal
→ In order traversal
→ Post order traversal.

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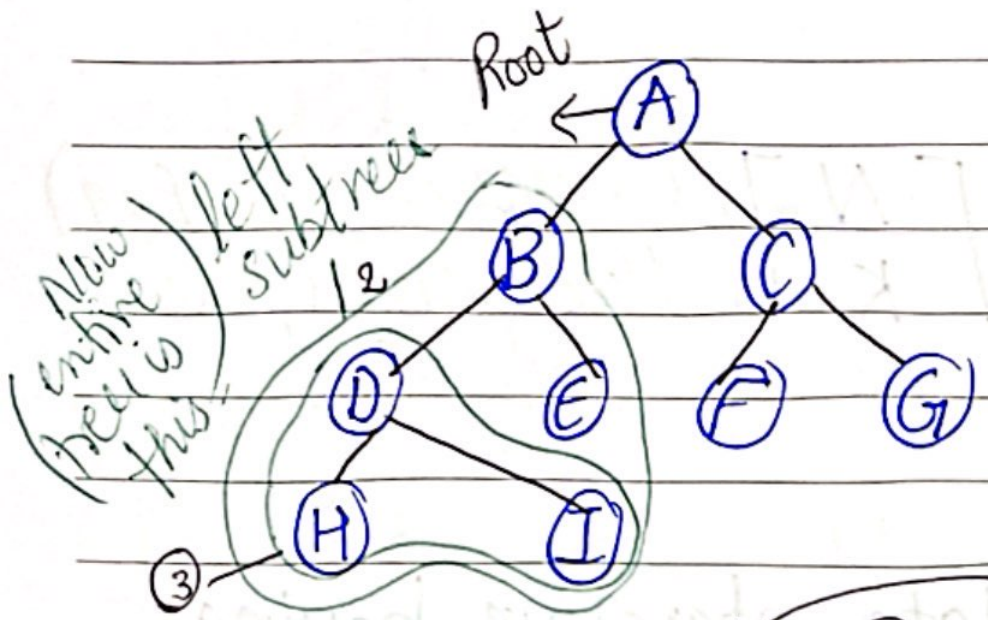
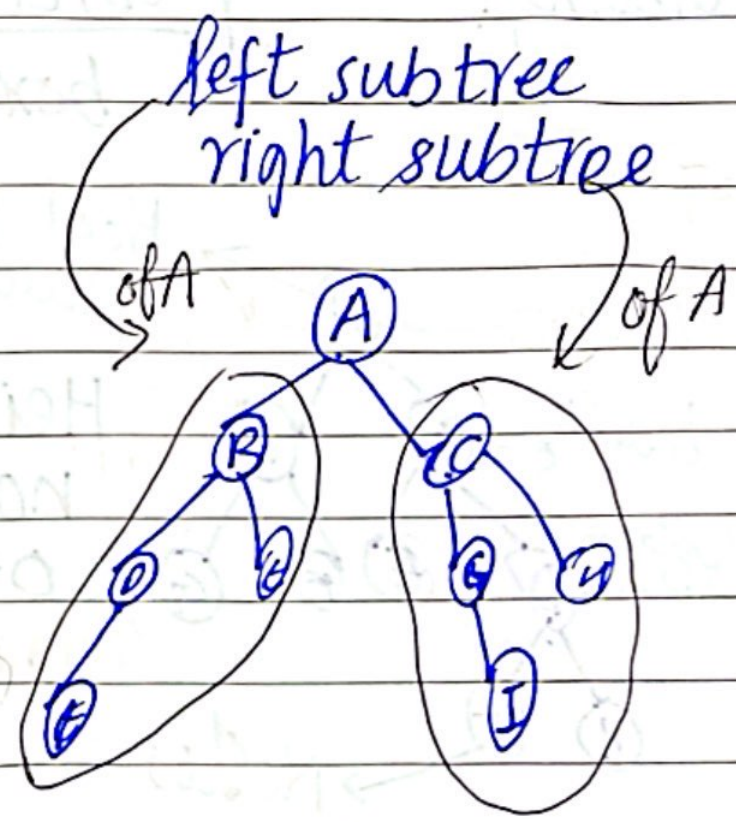
Pre - Order Traversal :



(Binary tree)



- Root is traversed (print)
- left subtree in pre-order

Call:

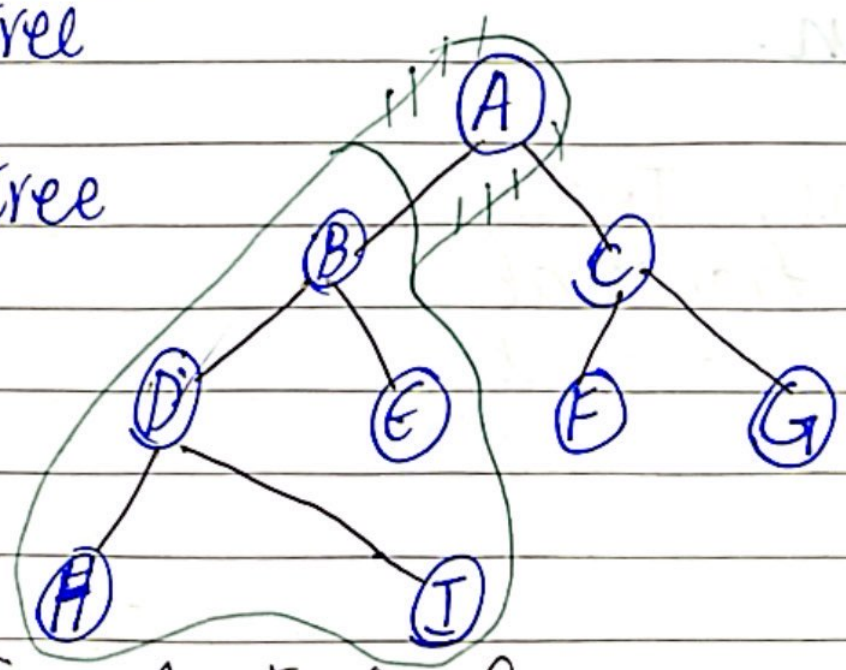
1) A, 2) B, 3) D, 4) H, 5) I, 6) E, 7) C, 8) F, 9) G.

- Right subtree in pre-order.

A, B, D, H, I, E, C, F, G.

In order Traversal.

- left sub-tree
- Root
- Right sub-tree



H, D, I, B, E, A, F, C, G.

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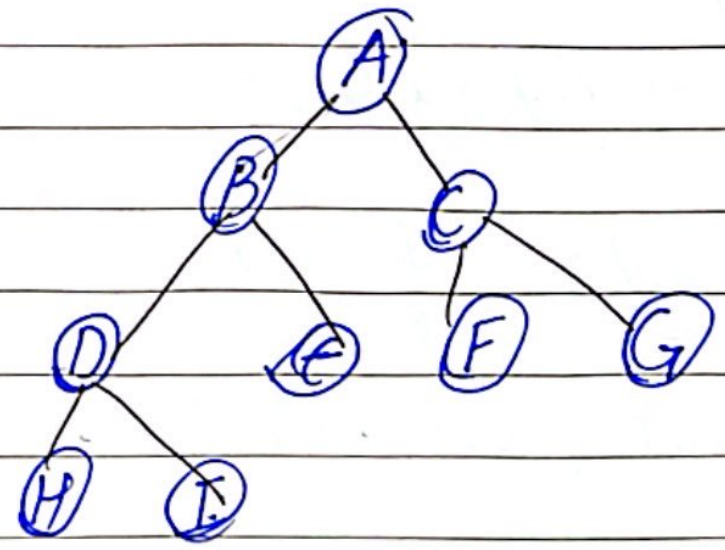
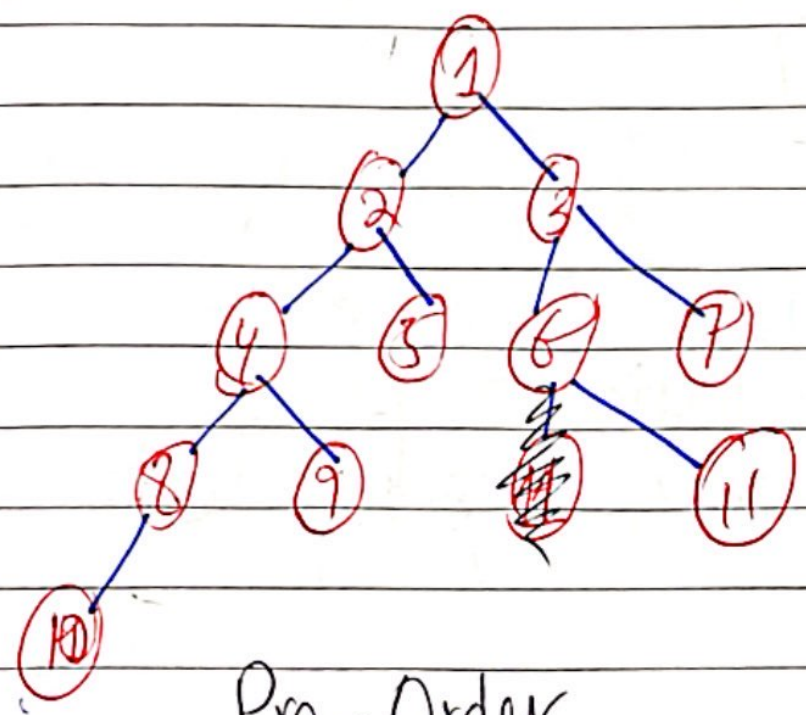
Post Order Traversal.

- Left
- Right
- Root



H, I, D, E, B, F, G, C, A

Apply all these.



In ~~Post~~-order

10, 8, 4, 9, 2, 5, 1, 6, 11, 3, 7

Post-Order.

10, 8, 9, 4, 5, 2, 11, 6, 7, 3, 1

Pre-Order.

1, 2, 4, 8, 10, 9, 5, 3, 6, 11, 7

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Relations & their Properties

→ sets (order pair)
→ members
→ members are related or not.

→ link of member of sets.

Cartesian Product

$A = \{1, 2, 3, 4\}$, $B = \{a, b, c\}$

$A \times B = \{ (1,a), (1,b), (1,c), (2,a), (2,b), (2,c), (3,a), (3,b), (3,c), (4,a), (4,b), (4,c), (4,d)$
(pairing)

Subset of $A \times B$ = Relation (R) = $\boxed{R \subseteq A \times B}$

Exercise:

Q1 List ordered pairs in the relation R from $A = \{0, 2, 1, 3, 4\}$ to $B = \{0, 1, 2, 3\}$ where $(a, b) \in R$ iff

b) $a + b = 4$

$R = \{ (a, b) \mid a + b = 4 \}$

$R = \{(1, 3), (2, 2), (3, 1), (4, 0)\}$

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divides

d) $a \mid b$

$R = \{(1,0), (2,0), (2,2), (3,0), (3,3), (4,0)\}$

greatest common divisor

e) $\gcd(a,b) = 1$.

$R = \{(0,1), (1,0), (1,1), (1,2), (1,3), (2,0)$ - - - - ..

Properties of Relation

- Reflexive (اپنے آپ سے connected ہو)
- Symmetric (new change کرنے کے بعد جگہ بدلنا چاہیے pair کی)
- Antisymmetric
- Transitive $(a,b) \in R$ and $(b,c) \in R$ then $(a,c) \in R$

Reflexive :

$B = \{0, 1, 2\}$ $\boxed{\text{every element should be reflexive.}}$

$R = \{(0,0), (1,1), (2,2)\}$ (reflexive)

$R = \{(0,0), (2,2)\}$ | $(1,1)$ missing so not reflexive.

Symmetric :

R $\boxed{\text{Any element can be symmetric}}$.

Examp: 7) R_2, R_3

Transitive :

Example 7: R_4, R_5, R_6 by default.

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Antisymmetric : (not symmetric)

R_4, R_5, R_6

$(a,b) \in R$ & $(b,a) \in R$ then $a = b$.

~~1 pair attached (not same) then change places then the thing come should not be there, that is anti-symmetric~~

Representing Relations Using matrices

Matrix : 2-D array
(0,1) matric
sai relation belong kr sktai hain.

i row
j coulumn.

m_{ij} : piece wise function.

$$m_{ij} = \begin{cases} 1 & \text{if } (a_j \cdot b_j) \in R \\ 0 & \text{if } (a_j \cdot b_j) \notin R \end{cases}$$

~~Example :01~~

After taking transpose and it come same then it is $\boxed{\text{symmetric}}$

If in centre 1 came it is $\boxed{\text{reflexive}}$.

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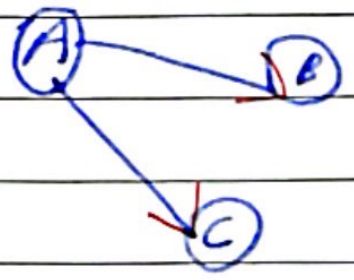
Representing Relation using graphs/directed graph.

- Graph : combination of vertices of edges
  plural of vertex

- loop.

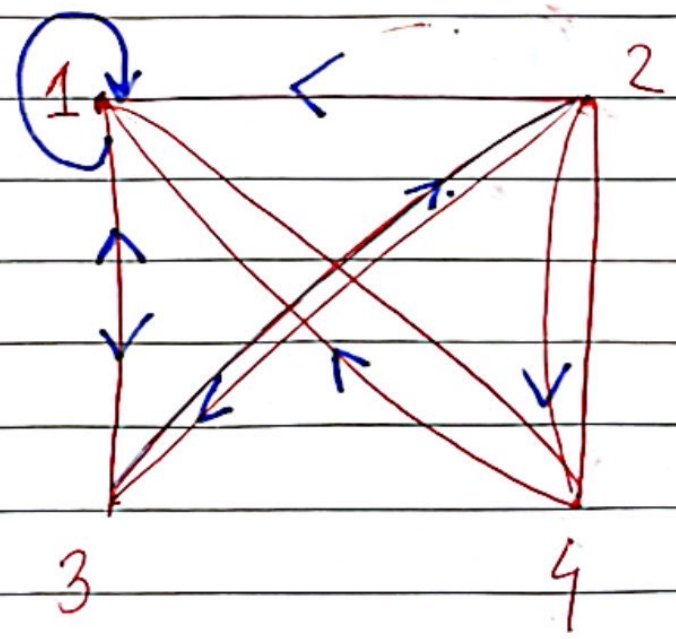


$\overrightarrow{AB}$
$\overrightarrow{AC}$
B and C are not directed graph.

Example 8

Directed graph of relation

$R = \{(1,1), (1,3), (2,1), (2,3), (2,4), (3,1), (3,2), (4,1)\}$



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